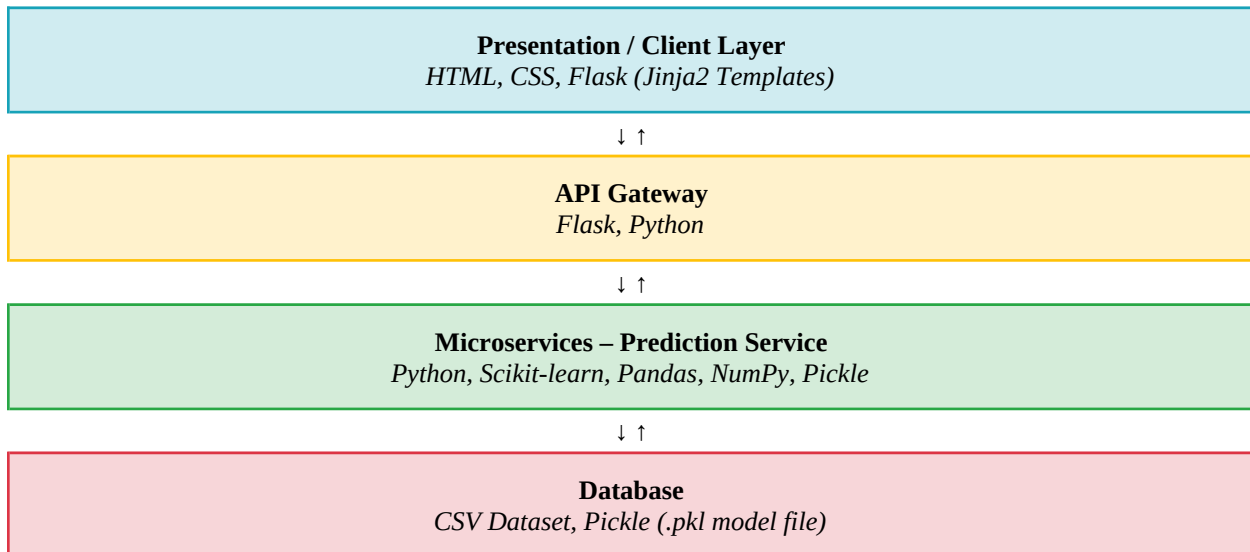


Solution Architecture

Date	15 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-being
Maximum Marks	2 Marks

Solution Architecture Diagram:

The diagram below shows the layered architecture of the HDI Predictor solution, from the Presentation Layer through the API Gateway and Prediction Microservice, down to the Database layer.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface where users enter HDI-related indicators (Life Expectancy, Expected Years of Schooling, Mean Years of Schooling, and GNI per Capita) and view the predicted Human Development Index and development category.	- HTML - CSS - Flask (Jinja2 Templates)
API Gateway	Receives user requests from the web interface, validates the input, forwards the data to the prediction model, and returns the prediction results to the user.	- Flask - Python
Microservices	Processes user input, loads the trained Machine Learning model, predicts the Human Development Index (HDI), classifies the development level (Low, Medium, High, or Very High), and sends the results	- Python - Scikit-learn - Pandas - NumPy

	back to the web application.	● Pickle
Database	Stores the HDI dataset used for training the Machine Learning model. The trained model is stored locally for prediction during application runtime.	● CSV Dataset ● Pickle (.pkl model file)